# UNIVERSITY OF THE PEOPLE

MATH 1280-01 - AY2026-T4

Assignment Activity Unit 1

Sampling Methods and Frequency Analysis

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### (a) Definitions with real-life example

Statistics is the science of collecting, organizing, analyzing, and interpreting data. Probability is the study of how likely an event is to occur. Population refers to the entire group of individuals or items that are being studied, while a sample is a smaller subset selected from the population for analysis.

For example, in a local café, if the owner wants to understand customer preferences, the population would be all customers visiting the café. A sample could be 50 randomly selected customers whose feedback is collected. Statistics would be used to analyze this feedback to understand preferences, while probability could help estimate the likelihood of customers choosing a specific drink.

### (b) Simple random sampling vs stratified sampling

Simple random sampling means every individual in the population has an equal chance of being selected. For example, selecting 50 café customers randomly from a complete customer list.

Stratified sampling divides the population into groups (strata) based on shared characteristics, then randomly selects from each group. For example, café customers could be divided into age groups (teenagers, adults, seniors), and samples are taken from each group to ensure balanced representation. This makes stratified sampling more accurate for diverse populations compared to simple random sampling.

## Question 2

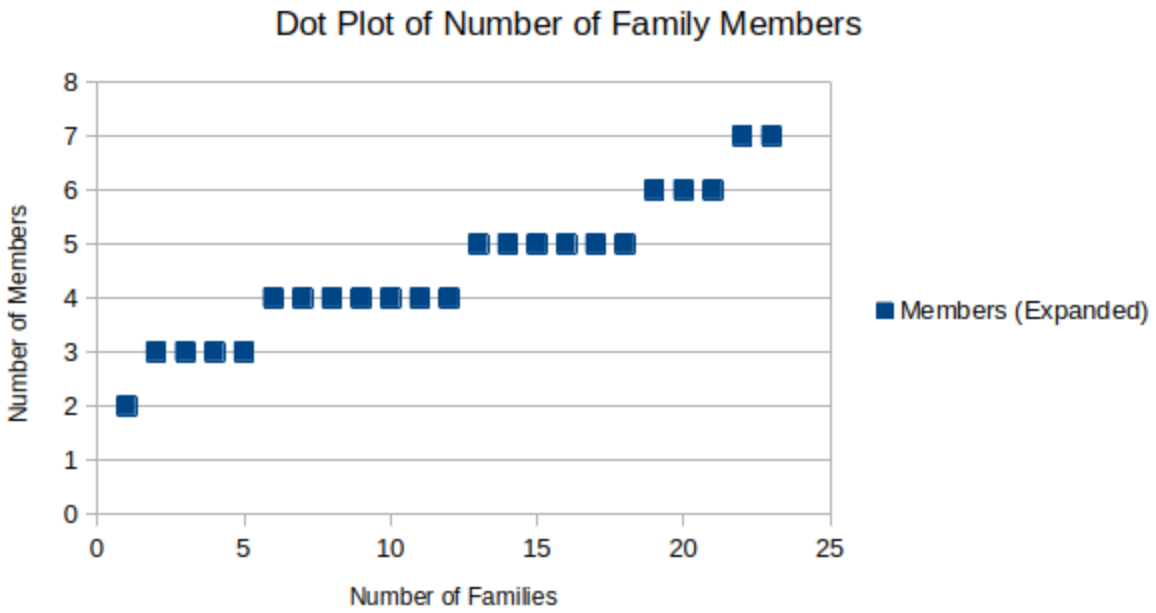
The given data represents the distribution of family members in a rural area.

### (a) Dot Plot

A dot plot was constructed to visually represent the distribution of family members across different households. Each dot in the plot represents one family.

- The horizontal axis (X-axis) shows the **Number of Families**
- The vertical axis (Y-axis) shows the **Number of Members**
- The chart is titled: **"Dot Plot of Number of Family Members"**

This plot helps in understanding how family sizes are distributed in the dataset.




## (b) Calculation of Families with More Than 4 Members

Families having more than 4 members include those with 5, 6, and 7 members.

- Families with 5 members = 6
- Families with 6 members = 3
- Families with 7 members = 2

**Total number of such families:**

6+3+2=11

## Final Answer:

The total number of families having more than 4 members is **11**.

# Question 3

The given table represents the cumulative frequency of hours studied by a group of students over a week.

### (a) Students who studied between 10 and 15 hours

To find the number of students in a specific interval, subtract the cumulative frequency of the previous interval from the cumulative frequency of the required interval.

Students (10–15 hours) = CF (0–15) − CF (0–10)
= 30 − 18
= 12

**Answer:** 12 students

### (b) Students who studied less than 20 hours

The number of students who studied less than 20 hours is directly given by the cumulative frequency of up to 20 hours.

Students (less than 20 hours) = CF (0–20)
= 45

**Answer:** 45 students

## References

Illowsky, B., & Dean, S. (2023). *Introductory statistics (2nd ed.)*. OpenStax, Rice University. https://openstax.org/details/books/introductory-statistics-2e

OpenStax. (2023). *Introductory statistics 2e textbook*. https://openstax.org